# Preliminary Project Report: GenAI Advisor, based on the Financial Advisor Bot template

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## Contents

**1 Introduction** **2**
1.1 Concept . . . 2
1.2 Motivation . . . 3
1.3 Scope . . . 3

**2 Literature Review** **4**
2.1 Generative AI in Financial Forecasting . . . 4
2.2 Explainable AI in Financial Systems . . . 4
2.3 Rule-based vs Machine Learning Investment Strategies . . . 5
2.4 Designing for Non-Technical Users in FinTech . . . 5
2.5 Backtesting and Evaluation in FinTech . . . 6
2.6 Synthesis and Research Gap . . . 6

**3 System Design** **8**
3.1 System Overview & Design Rationale . . . 8
3.2 Data Collection, Pre-processing & Feature Engineering . . . 9
3.3 Strategy Engine . . . 10
3.4 Explanation Generator . . . 11
3.5 User Interface . . . 12
3.6 Backtesting & Evaluation . . . 12

**4 Feature Prototype and Evaluation** **13**
4.1 Prototype Objective and Scope . . . 14
4.2 Implementation Details . . . 14
4.3 Evaluation Methodology . . . 15
4.4 Results and Comparison . . . 15
4.5 Discussion . . . 16
4.6 Conclusion . . . 16

# 1 Introduction

Artificial Intelligence (AI) continues to reshape global industries, with Generative AI (GenAI) standing out as one of the most influential recent developments. GenAI models, such as large language models (LLMs) and diffusion architectures, have driven widespread adoption and significant investment, particularly in the technology sector. Yet, for retail investors, the tools available to access this growth remain largely generic and inaccessible, both in terms of domain focus and interpretability.

Private equity and venture capital investments in GenAI startups are becoming larger and more targeted as investor strategies evolve with the technology. Funding in GenAI exceeded $56 billion in 2024, nearly double the $29 billion recorded in 2023 (14).

As shown in Figure 1 below, while the number of funding rounds declined in 2024, the overall deal value surged. This suggests a consolidation of capital into larger, infrastructure-oriented bets, aligning with investor confidence in scalable GenAI deployments. These trends highlight the need for tools that help investors evaluate publicly listed companies best positioned to benefit from this structural shift.

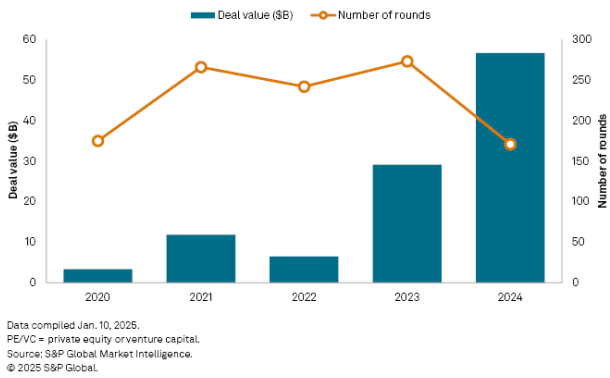


Figure 1: PE/VC investments in GenAI, 2020–2024. Source: S&P Global Market Intelligence (14).

This project responds to that gap by proposing the **GenAI Advisor**, a system that delivers explainable stock recommendations specifically for companies driving or enabling the GenAI revolution. Based on the *Financial Advisor Bot* template, the system is designed for non-technical users, providing accessible, transparent investment signals that go beyond traditional model outputs.

## 1.1 Concept

The GenAI Advisor is conceived as a personalised financial guidance tool for individual investors interested in the GenAI sector. It provides interpretable recommendations, whether to buy, hold, or sell specific stocks, alongside human-readable explanations for these decisions. Unlike traditional investment tools that obscure decision logic behind opaque algorithms or technical jargon, this system foregrounds explainability and domain relevance.

The intended users of this system are individual retail investors interested in emerging technology themes but lacking the technical background to interpret raw financial signals or operate sophisticated trading tools. These users are typically non-specialists with limited exposure to algorithmic finance or machine learning models. For this group, the GenAI Advisor offers an accessible decision-support tool that reduces complexity while preserving analytical rigour. It is also suitable for students, educators, or early-stage investors exploring explainable AI in financial contexts.

At its core, the GenAI Advisor is built on the belief that financial tools should be not only accurate but also comprehensible. This is particularly important in thematic investing, where

users are drawn not only by quantitative performance but by interest in specific technological trends. In this context, the system acts as a digital intermediary between complex market signals and users' intuitive understanding, offering advice that is grounded traceable, and aligned with a clearly scoped set of companies.

The conceptual novelty lies in its hybrid orientation: it combines evidence-driven financial analysis with a commitment to interpretability. The Advisor does not aim to outperform the market through high-frequency trading (HFT) or proprietary forecasting. Instead, it prioritises accessibility, domain focus, and trustworthiness, delivering insights users can understand and apply in their decision-making. In doing so, it invites users into the analytical process rather than substituting for it.

Furthermore, the GenAI Advisor embraces thematic integrity by carefully curating its stock universe. It avoids general-purpose technology firms in favour of those with demonstrable involvement in the GenAI ecosystem, whether through infrastructure, tooling, model development, or deployment. The rationale for company inclusion and exclusion is detailed in the project scope (Section 1.3). This conceptual grounding ensures that recommendations are not only technically derived but also strategically aligned with the user's thematic intent.

## 1.2 Motivation

The motivation for this project is both technical and user-oriented. On one hand, GenAI represents an emerging thematic investment opportunity underserved by existing tools. On the other, there is a growing demand for transparent, user-friendly AI systems in finance.

Traditional robo-advisors, such as Nutmeg and Moneyfarm, provide automated portfolio allocation but are domain-agnostic and opaque in how decisions are made. While open-source trading bots like Freqtrade offer flexibility, they require technical expertise and lack natural language interfaces. StockGPT (9) demonstrates the potential of generative models in stock prediction, while recent work on XAI-integrated forecasting (10) highlights the importance of trust and interpretability in financial AI systems.

By combining simple rule-based logic with predictive models and natural language explanations, the GenAI Advisor aims to deliver recommendations that are not only accurate but also intelligible. A local LLM serves as an interpretive layer rather than a decision engine, ensuring that outputs remain grounded in logic and statistical inference, while still being comprehensible to the user.

Beyond usability, the project contributes to responsible AI by ensuring data privacy and reproducibility. No personal or sensitive data is collected, and all sources are public. The LLM is hosted locally to avoid transmitting user inputs externally. The system is explicitly intended for educational and demonstrative use, and includes appropriate disclaimers to distinguish it from regulated financial services.

## 1.3 Scope

The GenAI Advisor focuses on a curated set of publicly traded companies whose business models are materially driven by Generative AI. The portfolio includes firms across three categories: (1) software developers and strategic investors (e.g., Microsoft, Alphabet, Meta, Amazon); (2) hardware providers (e.g., Nvidia, AMD, Intel, Marvell); and (3) infrastructure suppliers (e.g., TSMC, SK hynix, Broadcom, Arista Networks). These companies were selected for their direct contributions to GenAI model development, deployment, or enablement.

To maintain thematic focus and ensure data availability, the system excludes private companies, indirect holdings, and general technology firms lacking core GenAI offerings (e.g., Apple, Netflix). Chinese firms (e.g., Alibaba, Tencent, Baidu) are also excluded due to market access limitations (16) and restricted financial transparency due to the Variable Interest Entity (VIE) structure of the BATX (Baidu, Alibaba, Tencent, and Xiaomi) stocks (4).

This scope ensures the system operates within a clearly bounded, analytically defensible domain. It enhances the relevance of recommendations by aligning them with user intent and facilitates reproducible evaluation by focusing on transparent, publicly accessible data.

# 2 Literature Review

The rapid advancement of Generative AI (GenAI) has sparked significant interest in its application to financial forecasting and investment support systems. Despite notable progress in predictive modelling, current financial advisory tools often prioritise performance metrics at the expense of transparency and interpretability. This trade-off presents a particular challenge for non-technical retail investors, who require not only accurate but also comprehensible recommendations in order to make informed decisions.

This literature review examines recent developments at the intersection of GenAI, explainable AI (XAI), and algorithmic trading. It explores generative models for financial prediction, the integration of XAI techniques in forecasting systems, and the implications of user-centred design in FinTech. The aim is to critically evaluate existing approaches and identify a gap in the current landscape: the lack of accessible, domain-specific, and explainable investment tools. The review concludes by outlining how the *GenAI Advisor* project is designed to address this gap through a hybrid architecture that integrates transparent rule-based logic, supervised machine learning, and natural language explanation layers.

## 2.1 Generative AI in Financial Forecasting

The application of Generative AI (GenAI) in financial domains has emerged as a frontier area of research, particularly in the context of algorithmic trading and decision support. While traditional predictive systems rely on time-series models and supervised learning techniques, recent work has demonstrated the viability of large language models (LLMs) and generative transformers for tasks such as stock sentiment analysis, earnings prediction, and trade signal generation.

A notable example is *StockGPT* by Mai (9), which applies a fine-tuned generative language model to extract predictive signals from financial text and historical data. The system shows promising results in backtesting but offers limited transparency in terms of feature attribution and decision rationale. This highlights an emerging trade-off in GenAI systems between performance and interpretability, particularly in high-stakes domains like finance.

Generative models have also been used to simulate synthetic market data (15), enabling improved model training under data scarcity. However, these approaches often lack integration with rule-based or statistical baselines, making it difficult to assess the marginal utility of the generative component. Furthermore, the absence of human-centred explanation interfaces in these systems poses a barrier to adoption among non-specialist users.

These limitations underline the need for hybrid architectures that balance the predictive power of generative models with the interpretability of traditional techniques. The *GenAI Advisor* addresses this by separating signal generation from explanation, thereby ensuring that the core decision logic remains auditable while leveraging GenAI to enhance accessibility and user trust.

## 2.2 Explainable AI in Financial Systems

As AI-driven systems increasingly influence financial decision-making, the demand for explainable artificial intelligence (XAI) has become a critical concern. In domains characterised by uncertainty, regulatory oversight, and end-user scepticism, opaque "black-box" models are often ill-suited for practical deployment (6). This is particularly true for retail investors who may lack technical backgrounds yet require justifiable, transparent investment reasoning.

Marey et al. (10) propose an empirical framework that integrates deep learning with local explanation techniques, such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations). Their study demonstrates that user trust and engagement are positively correlated with the presence of intelligible model rationales, especially when users are allowed to compare multiple explanation formats. However, these explanation mechanisms are typically appended to existing models rather than embedded in the design, leading to potential inconsistencies between what the model decides and what is communicated.

Other work, such as Ribeiro et al. (12), highlights the trade-off between global accuracy and local interpretability. While LIME-style approximations offer useful insight into model behaviour, they can misrepresent model logic under certain conditions, particularly in non-linear financial regimes.

To address these issues, the *GenAI Advisor* integrates XAI at both the structural and presentation layers. Rule-based logic provides a baseline of transparent, auditable recommendations, while machine learning classifiers offer enhanced signal detection. A local language model translates system outputs into natural-language explanations, preserving alignment between system behaviour and user-facing justifications. This layered approach ensures both decision accuracy and explanatory coherence.

### 2.3 Rule-based vs Machine Learning Investment Strategies

Investment recommendation systems have traditionally relied on rule-based logic grounded in technical indicators such as moving averages, momentum oscillators, and price-volume trends. These heuristics offer a high degree of interpretability, making them attractive to both practitioners and novice investors (11). Strategies based on conditions like moving average crossovers or Relative Strength Index (RSI) thresholds are easily understood and lend themselves to direct explanation.

However, rule-based systems often fail to capture the non-linearities and temporal dependencies inherent in financial time series. This has led to a growing interest in machine learning (ML) models such as Random Forests, Gradient Boosting Machines (e.g. XGBoost), and neural networks for stock selection and signal generation (5). These models can discover subtle patterns and adapt to changing market regimes, offering superior predictive performance in some contexts.

Yet, the adoption of ML in retail investment contexts is hindered by concerns over transparency and robustness. Ryll and Seidens (13) highlight that black-box models frequently suffer from overfitting, poor generalisation, and limited interpretability when applied to financial data. Chakraborty and Joseph (3) similarly observe that financial institutions remain cautious about deploying ML in high-stakes environments due to its opacity and the risk of unpredictable behaviour under changing market conditions. These issues are especially problematic for non-expert users, who rely on trustworthy and intelligible decision support systems.

The *GenAI Advisor* bridges this gap by employing a hybrid strategy engine. Transparent rules serve as a reliable baseline, while supervised learning models (Random Forest and XGBoost) are used to enhance signal strength by capturing complex interactions. This dual approach enables comparative evaluation and supports user trust, as each recommendation is either traceable to a logical rule or supported by model-driven evidence.

### 2.4 Designing for Non-Technical Users in FinTech

The usability and accessibility of financial technologies remain a significant challenge, particularly for non-technical users who may lack domain expertise in data science or quantitative trading. FinTech interfaces that prioritise raw performance often neglect explainability and user-centred design, which are essential for effective decision support and adoption among retail investors. Ben David et al. (2) demonstrate through experimental evidence that the inclusion

of explainable AI elements significantly improves trust and adoption rates among users of financial algorithmic advisors. Their findings underscore the critical role of intelligibility and transparency in interface design for enhancing user engagement.

Studies have shown that users without a financial background exhibit higher levels of anxiety and lower confidence when interacting with complex or opaque recommendation systems. Without clear explanations, users may disregard otherwise effective recommendations, especially in high-stakes settings involving personal investments. This issue is compounded by the fact that many retail-focused platforms do not disclose the reasoning behind their advice, instead presenting deterministic outcomes with minimal context (17).

Explainable user interfaces (XUIs) have been proposed as a solution, incorporating elements such as visual signal annotation, confidence metrics, and natural-language summaries (2). These approaches have demonstrated success in increasing perceived trust, interpretability, and willingness to act upon AI-generated recommendations. However, implementation of such systems is often inconsistent, and few are tailored to specific thematic portfolios like those in GenAI-focused sectors.

The *GenAI Advisor* responds to this gap by embedding a natural language explanation layer that interprets rule-based and ML-driven signals into plain-English rationales. By adopting a language-first, model-agnostic approach, the system enables users to engage meaningfully with its outputs, fostering transparency and informed decision-making without requiring technical fluency.

### 2.5 Backtesting and Evaluation in FinTech

Robust evaluation is a critical component of financial algorithm design, ensuring that investment strategies are not only theoretically sound but also empirically validated under realistic market conditions. Backtesting—retrospective testing of a strategy using historical data—remains the standard approach for evaluating financial models. However, its reliability depends on careful control for biases such as lookahead error, data snooping, and overfitting (1).

Evaluation metrics in the FinTech domain extend beyond accuracy to include financial risk-return measures. The Sharpe Ratio, for instance, assesses risk-adjusted return by penalising volatility, while maximum drawdown quantifies peak-to-trough losses—both essential for comparing the robustness of competing strategies (8). Other common metrics include total return, alpha and beta coefficients, and classification-based indicators such as precision, recall, and F1-score when using ML classifiers for signal prediction.

Tools such as `backtrader` and `pandas`-based frameworks facilitate integration of financial indicators, portfolio simulation, and metric reporting, making them essential in contemporary algorithmic finance development. Nevertheless, evaluation practices remain inconsistent, with many academic and commercial models reporting high in-sample performance without adequate out-of-sample or cross-validation testing (7).

The *GenAI Advisor* explicitly incorporates backtesting into its pipeline using reproducible Python-based workflows. It compares rule-based and machine learning strategies across multiple indicators and time frames, reporting both financial and statistical performance metrics. This dual-layered evaluation provides transparency and supports empirical justification for investment recommendations.

### 2.6 Synthesis and Research Gap

The review of existing literature reveals a rich landscape of innovation at the intersection of AI, finance, and user-centred design. However, it also exposes clear limitations in current approaches when evaluated through the lens of accessibility, explainability, and domain specificity—three attributes that are critical for supporting non-technical investors seeking exposure to GenAI-focused equities.

Generative models such as *StockGPT* demonstrate that large language models (LLMs) can be fine-tuned to extract financial signals from unstructured data, yet their outputs often lack interpretability and alignment with human-understandable trading logic (9). While this presents a promising avenue for performance gains, it simultaneously creates a transparency deficit. Similarly, Takahashi and Mizuno (15) propose using diffusion models to generate synthetic financial time series that replicate complex market behaviours. Although valuable for addressing data scarcity and irregularities, these techniques rarely incorporate explanatory mechanisms or support integration with domain-informed baselines, limiting their applicability in systems designed for human-in-the-loop decision making.

Research on XAI has provided a broad array of model-agnostic tools such as SHAP and LIME (12), which can help reveal internal model dynamics. Yet, as highlighted by Marey et al. (10), these methods are frequently bolted onto systems post hoc and suffer from a mismatch between model behaviour and user comprehension. Additionally, the form and delivery of explanations have not been standardised across FinTech applications, leaving a usability gap that affects trust and adoption.

This concern is amplified when considering ML-only investment systems. Despite their performance advantages, such models are often developed without sufficient consideration for interpretability or operational auditability. Ryll and Seidens (13) highlight that machine learning models in financial forecasting frequently suffer from overfitting and poor generalisation, particularly under changing market regimes. Similarly, Chakraborty and Joseph (3) caution that opaque model architectures pose a challenge for real-world deployment in financial institutions, where accountability and stability are essential. Conversely, rule-based systems offer clarity and transparency but often lack the adaptability required to navigate complex, non-linear market dynamics.

From a usability perspective, research indicates that FinTech platforms frequently underestimate the cognitive demands placed on users, often failing to present decision rationales in a digestible and actionable format. Ben David et al. (2) demonstrate that the inclusion of explainable elements—such as natural-language justifications and transparency cues—significantly improves user trust and the likelihood of adopting AI-generated financial advice. While some interfaces include features like confidence bands or risk scores, few extend to personalised, intelligible explanations grounded in model logic and user context. Moreover, no existing platform explicitly addresses the needs of investors seeking targeted exposure to the GenAI sector, a growing interest area underserved by conventional robo-advisors.

In terms of evaluation, there is a consistent pattern of insufficient out-of-sample validation and limited comparative benchmarking across strategy types (1; 7). Financial metrics such as Sharpe Ratio or drawdown are inconsistently applied, and many studies prioritise classification accuracy without accounting for investor risk tolerance or decision thresholds.

Taken together, these findings point to a clear research and development gap: there is currently no publicly available tool that offers GenAI-specific stock recommendations through a hybrid system combining auditable rules, machine learning models, and explainable outputs tailored to non-expert users. Moreover, there is a lack of reproducible, modular pipelines that support comparative evaluation of rule-based and ML-based strategies in the context of GenAI sector investment.

The *GenAI Advisor* aims to fill this gap by providing an integrated, open-source prototype that embodies the insights drawn from this literature. It does so through: (1) a hybrid strategy engine that combines clarity and complexity; (2) a local LLM explanation layer to generate user-friendly rationales; (3) focused stock screening based on GenAI sector relevance; and (4) a rigorous evaluation framework using both financial and statistical metrics. By unifying these components, the system advances the current state of practice in explainable FinTech design for thematic investing.

# 3 System Design

This chapter presents a detailed account of the design of the GenAI Advisor system, organised around a five-layer architecture: data collection and feature engineering, strategy engine, explanation generator, user interface, and backtesting and evaluation. The system is engineered to deliver explainable, contextual, and user-friendly financial insights for retail investors with an interest in generative AI equities.

## 3.1 System Overview & Design Rationale

The architecture of the GenAI Advisor system is designed to support explainability, modularity, and extensibility across all components. It is composed of five distinct yet interdependent layers:

1. **Data Collection, Pre-processing & Feature Engineering** — responsible for sourcing, cleaning, and transforming financial data into formats suitable for analysis and embedding.
2. **Strategy Engine** — the analytical core that integrates retrieved knowledge and feature representations to generate structured investment insights.
3. **Explanation Generator** — enhances transparency by producing traceable justifications and evidence-backed rationales for each recommendation.
4. **User Interface** — delivers a conversational, intuitive interaction point for retail users, visualising results in a digestible and accessible format.
5. **Backtesting & Evaluation** — supports validation of generated recommendations through historical simulations and user feedback loops.

Each layer encapsulates a single responsibility while remaining interoperable with adjacent layers, reflecting a separation-of-concerns design philosophy. This modular approach enables independent development and testing of each component, reducing coupling and simplifying future iterations or extensions.

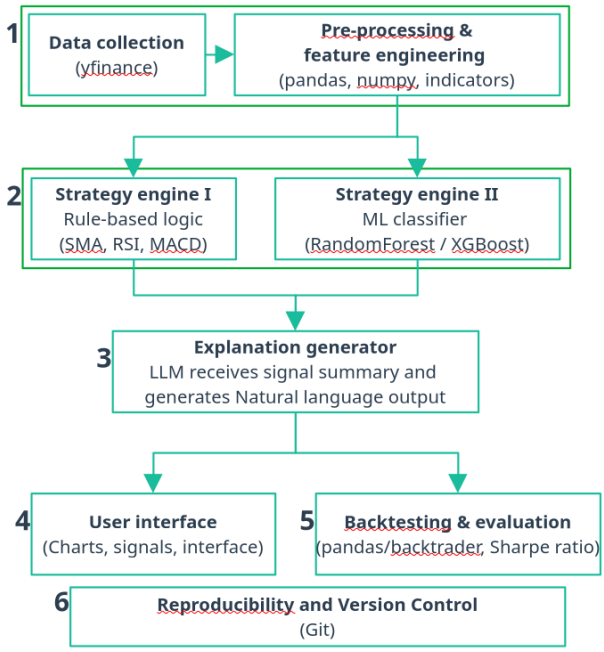


Figure 2: Five-layer Architecture of GenAI Advisor

The design rationale also reflects broader system requirements, particularly the need for explainability, adaptability, and integration with real-time financial data streams. Embedding traceability into the architecture ensures that every recommendation is grounded in verifiable data sources, supporting responsible AI usage. Furthermore, by situating the explanation generator as a distinct layer, the architecture highlights the importance of user trust and transparency as core design principles.

Scalability is addressed through stateless API services, containerised deployment, and asynchronous task handling, enabling the system to scale with increasing data volume and user interactions. The architecture is cloud-agnostic, allowing deployment on various infrastructure backends, and is compatible with both proprietary and open-source LLMs.

This five-layer design positions the GenAI Advisor as a robust, auditable, and user-centred solution for explainable financial intelligence.

### 3.2 Data Collection, Pre-processing & Feature Engineering

This foundational layer is responsible for acquiring, preparing, and transforming financial data into structured, analysable formats that support both retrieval and modelling. It ensures that the inputs to the system are reliable, consistent, and semantically rich.

**Data Sources:** Financial data is sourced via the `yfinance` Python library, which extracts structured data from Yahoo Finance. This includes:

- Daily OHLCV (open, high, low, close, volume) price data
- Adjusted close prices accounting for dividends and splits
- Market fundamentals including P/E ratio, EPS, market capitalisation, and beta
- Dividend history, earnings results, and analyst forecasts
- Corporate actions such as stock splits and ex-dividend dates

**Pre-processing:** All raw data is immediately pre-processed in memory using `pandas` and associated libraries before being persisted. This includes:

- Handling of missing values and filtering of anomalous entries
- Normalisation and transformation of numerical columns (e.g. z-score, log returns)
- Resampling and forward-filling to align time indices across features
- Annotating rows with relevant metadata (e.g. ticker symbol, sector classification, timestamps)

This pre-processing ensures consistency across all instruments and time periods and reduces computational load in downstream stages.

**Data Storage:** Once cleaned and transformed, the processed dataset is stored in a relational MariaDB database. This allows for fast, structured access by subsequent system components and avoids repeated API calls to Yahoo Finance. Tables are designed to support efficient filtering by date, ticker, and feature type. Indexes are maintained on primary keys and time columns to optimise query latency.

**Feature Engineering:** This stage derives higher-order signals and representations from the cleaned data:

- **Technical indicators:** including moving averages, MACD, RSI, and volatility bands
- **Fundamental deltas:** e.g. growth rates in earnings, changes in valuation multiples

- **Risk signals:** calculated from rolling variance, beta, and return dispersion
- **Embeddings:** tabular records are converted into dense vector representations using Sentence-BERT for semantic similarity and retrieval

**Database Schema:** The MariaDB schema consists of the following core tables:

- `daily_prices(ticker, trade_date, open, high, low, close, adj_close, volume)`
- `company_fundamentals(ticker, name, sector, industry, market_cap, pe_ratio, eps, beta, dividend_yield, last_updated)`
- `technical_indicators(ticker, trade_date, sma_20, sma_50, rsi_14, macd, bollinger_upper, bollinger_lower)`
- `earnings_calendar(ticker, earnings_date, expected_eps, reported_eps, surprise_pct)`

These tables are indexed on `ticker` and `date`-related fields to support fast retrieval and filtering during downstream analysis.

**Design Justification:** The choice of MariaDB for structured data storage reflects a balance between transparency, performance, and system complexity. Relational databases offer well-defined schema enforcement, support for indexed queries, and seamless integration with Python data analysis libraries. This aligns with the project's emphasis on explainability, traceability, and reproducibility. Alternative solutions were considered, including DuckDB for embedded analytics, Parquet-backed storage for columnar efficiency, and TimescaleDB for specialised time-series features. However, these options introduce either added complexity or are optimised for larger-scale, concurrent systems. For the scope of this project—focusing on iterative prototyping, backtesting, and clarity—MariaDB provides a robust and pedagogically sound foundation.

### 3.3 Strategy Engine

The strategy engine forms the analytical core of the GenAI Advisor system, responsible for evaluating investment opportunities using structured signals and domain-specific rules. It interprets features derived from historical data and generates intermediate outputs such as buy/sell flags, risk scores, and confidence levels. This layer operates independently of any generative models and is designed for transparency, reproducibility, and interpretability.

**Signal Retrieval:** At runtime, the strategy engine queries the MariaDB backend for time-aligned features such as technical indicators, valuation metrics, volatility profiles, and momentum scores. Data is filtered by ticker, date, and sector using predefined criteria. Missing values are handled via forward-filling or exclusion depending on the indicator's sensitivity.

**Rule-based Scoring:** Initial decision logic is implemented using deterministic rules. For example:

- *Momentum signal:* Generate a buy signal if the closing price exceeds the 50-day simple moving average and RSI is in the range 50–70.
- *Valuation screen:* Flag as overvalued if the P/E ratio is greater than the sector median by more than one standard deviation.
- *Risk classification:* Assign a "high risk" label if beta exceeds 1.5 or if recent earnings variance is above a defined threshold.

These rule sets are configurable and grounded in established financial heuristics.

**Machine Learning Models:** Where applicable, supervised models such as logistic regression or gradient-boosted decision trees may be used to estimate directional signals. Input

features include moving averages, volatility, return profiles, and sector-adjusted valuation ratios. Outputs are binary or probabilistic classifications of investment suitability.

**Strategy Output:** The final output is a structured intermediate recommendation, for example:

- *Ticker:* MSFT
- *Action:* HOLD
- *Confidence:* 72%
- *Tags:* "Moderate Risk", "Overvalued"

This output is passed to the explanation layer, where it is contextualised and communicated to the end-user through a natural language interface.

The strict separation of decision logic from explanation ensures that recommendations are interpretable, reproducible, and auditable across time.

### 3.4 Explanation Generator

The explanation generator is responsible for converting structured strategy outputs into human-readable narratives. This layer interfaces with a large language model (LLM) to articulate the rationale behind each recommendation, ensuring the system remains explainable, transparent, and aligned with the needs of non-expert retail investors.

**Prompt Construction:** Each recommendation generated by the strategy engine is converted into a structured prompt. This prompt contains:

- The recommended action (e.g. BUY, HOLD, SELL)
- Associated features (e.g. P/E ratio, RSI, volatility)
- Descriptive tags (e.g. "Overvalued", "Moderate Risk")
- Ticker metadata (e.g. sector, market cap, recent price movement)

The prompt template enforces a consistent narrative structure and ensures that only relevant, pre-computed data is surfaced to the LLM. No raw retrieval or document inference is performed at this stage.

**LLM Invocation:** The prompt is submitted to a locally hosted language model, with inference restricted to models that can be executed on a GPU with 8GB of VRAM. This design constraint promotes offline operability, cost control, and reproducibility. Candidate models include distilled or quantised variants of open-source architectures such as Mistral, LLaMA, or Phi-2.

**Post-processing and Attribution:** The LLM output is parsed and linked to its underlying features. Numerical drivers are tagged and cross-referenced with the original strategy inputs to enable traceability. Where applicable, confidence indicators and relevant feature values are surfaced alongside the textual response.

**Model Selection Rationale:** For the purpose of this project, fine-tuning large language models is deliberately excluded due to its high resource demands and limited added value in the context of structured explanation generation. Instead, the system leverages pre-trained models served locally via the Ollama framework, which supports quantised variants of open-source architectures such as Mistral, Phi-2, and LLaMA2. These models strike a practical balance between performance and hardware feasibility, running comfortably on an 8GB VRAM GPU. The focus is placed on prompt design and response evaluation rather than model adaptation. This strategy aligns with the project's emphasis on transparency, reproducibility, and rapid

iteration, while avoiding the complexity, cost, and irreproducibility risks associated with custom model training.

By isolating the explanation function from decision-making logic and constraining it to local, auditable models, this design ensures the advisory system remains intelligible, trustworthy, and compliant with practical deployment constraints.

### 3.5 User Interface

The user interface (UI) serves as the primary point of interaction between the end-user and the advisory system. It is designed to surface system outputs in a clear, concise, and context-aware format while abstracting away the technical complexity of the underlying architecture. The target user is a retail investor with limited technical background but an interest in gaining structured insight into generative AI equities.

**Interface Design Principles:** The UI is guided by the principles of usability, transparency, and minimalism. It aims to provide information density without cognitive overload, offering progressive disclosure of details. The design supports both exploratory queries and structured evaluations.

**Interaction Model:** The primary mode of interaction is a natural language query interface, where users submit questions such as "Is NVIDIA a good investment right now?" or "Which GenAI stocks show earnings growth?". Queries are routed through the backend pipeline and responses are returned in structured, explanation-rich format.

**Presentation Layer:** Output is displayed in modular response cards, with each card corresponding to a company. Each card includes:

- The recommended action (BUY, HOLD, SELL)
- A summarised rationale generated by the explanation layer
- Key features (e.g. P/E ratio, RSI, beta) highlighted inline
- Option to expand for full explanation and numerical context

Cards are sorted based on confidence or relevance, and users can collapse or pin cards for comparison.

**Technical Implementation:** The frontend is built using a lightweight web framework such as Streamlit or Flask with React components. This ensures rapid prototyping, cross-platform compatibility, and support for secure API calls to backend services. The UI is deployed as a local or server-hosted web application accessible via browser.

**Responsiveness and Accessibility:** The interface is optimised for both desktop and mobile devices. Layouts use responsive containers and typography is chosen for readability. Accessibility guidelines such as contrast ratios and semantic labelling are followed to ensure inclusivity.

This interface completes the user-facing layer of the system, enabling seamless communication of insights derived from structured analysis and local language model generation.

### 3.6 Backtesting & Evaluation

This layer is responsible for assessing the performance, consistency, and interpretability of the GenAI Advisor system. Evaluation is conducted along two axes: (i) financial effectiveness of the recommendations and (ii) fidelity and stability of the generated explanations.

**Backtesting of Strategy Output:** The system supports retrospective evaluation of strategy outputs using historical financial data stored in MariaDB. Given a time window and a set of tickers, the engine replays strategy decisions based on data available at the time. The simulated trades are benchmarked using standard performance metrics such as:

- Cumulative return
- Sharpe ratio
- Maximum drawdown
- Hit rate (proportion of correct directional calls)

These simulations validate whether the rule-based and ML-driven components produce economically viable signals under realistic conditions.

**Evaluation of Explanation Consistency:** Because the system includes a generative component, it is essential to assess the consistency and fidelity of explanations. For a given strategy output, repeated LLM invocations are evaluated for semantic agreement and factual stability. Techniques include:

- *Stability under identical prompts:* The same input is submitted multiple times to check for drift.
- *Paraphrase testing:* Slightly altered prompts are compared for explanation robustness.
- *Key fact alignment:* Extracted metrics in the explanation are compared against known strategy outputs.

These tests ensure that the narrative remains faithful to the underlying numerical reasoning.

**User Feedback Loop:** The UI logs user reactions to recommendations through binary feedback (e.g. thumbs up/down) and optional free-text comments. Feedback is stored with associated prompt and strategy metadata, enabling post-hoc review and potential retraining of classification thresholds or strategy weights.

**Qualitative Auditing:** In addition to automated tests, a sample of responses is manually audited to assess:

- Clarity of the natural language explanation
- Justification coverage (i.e., do the reasons align with the action?)
- Absence of hallucinated content or misleading terminology

This step supports ethical AI principles by ensuring outputs remain comprehensible and appropriate for retail consumption.

This evaluation layer closes the loop between system logic, narrative explanation, and end-user experience. It provides the foundations for continuous improvement while maintaining interpretability and trust.

# 4 Feature Prototype and Evaluation

This section presents a focused implementation of the explanation generator and a comparative evaluation of several local large language models (LLMs). The objective is to determine which model best aligns with the system's requirements for explainability, resource efficiency, and accessibility. The evaluation is scoped to models runnable within an 8 GB VRAM constraint to support reproducibility and local deployment.

## 4.1 Prototype Objective and Scope

The prototype isolates the final layer of the GenAI Advisor architecture—the explanation generator. It takes as input a structured recommendation, typically generated by the strategy engine, and produces a textual rationale. This explanation layer is central to the system's commitment to transparency and user trust. For the purposes of this prototype, upstream stages such as data collection, feature engineering, and scoring logic are simulated using manually specified inputs. This allows a controlled environment in which model outputs can be directly compared.

The task is framed as follows: given a set of financial signals and a recommendation (e.g., BUY, HOLD, SELL), can a language model explain the recommendation in a manner that is accurate, clear, and suitable for a non-technical retail investor? The success of the explanation layer depends on the model's ability to infer causality or justification purely from the structured metrics it receives.

## 4.2 Implementation Details

The prototype is implemented in Python and interfaces with the `Ollama` runtime to serve quantised language models locally. Ollama is selected for its compatibility with consumer-grade hardware and its support for a growing list of well-maintained, open-source models. Using this framework, models can be served from the command line or accessed via HTTP API for more structured interaction.

All models tested are quantised 7B variants, chosen to balance performance with hardware constraints. Models include:

- `mistral-7b` — A strong general-purpose instruction-tuned model known for its coherence and performance on reasoning tasks.
- `phi-7b` — Optimised by Microsoft for compactness and low hallucination rates.
- `gemma-7b` — Released by Google, this model offers efficient inference and strong factual alignment.
- `llama2-7b` — Meta's widely adopted model, used here as a baseline.
- `deepseek-r1-7b` — A reasoning-oriented model with good handling of structured data.
- `vicuna-7b`, `codellama-7b`, `starling-lm-7b`, `neural-chat-7b`, and others — evaluated for comparative performance.

Each model is presented with a structured prompt template. No commentary or text is provided; the model must infer and generate all rationale. For example:

```
You are a financial assistant. Based on the following metrics:
Ticker: MSFT
Action: BUY
P/E Ratio: 32.1
RSI: 58
Beta: 1.1

Explain in simple terms to a layperson why this action is appropriate, using only the signal
```

Additional test prompts used during evaluation include:

```
You are a financial assistant. Based on the following metrics:
Ticker: NVDA
Action: HOLD
P/E Ratio: 75.6
RSI: 69
Beta: 1.8

Explain the rationale behind this HOLD recommendation to a retail investor.
```

```
You are a financial assistant. Based on the following metrics:
Ticker: AMZN
Action: SELL
P/E Ratio: 92.4
RSI: 73
Beta: 1.4

Explain in accessible language why this stock may no longer be suitable to hold.
```

Each of these prompts reflects a different combination of quantitative signals to test the model's consistency and interpretive ability across scenarios.

### 4.3 Evaluation Methodology

Evaluation focuses on three core criteria:

1. **Factual Alignment**: Whether the explanation references and reflects the correct input signals (e.g., "high P/E" when P/E exceeds 80).
2. **Interpretability**: Whether the rationale is readable, free from jargon, and suitable for a non-expert.
3. **Stability**: Whether the same prompt produces consistent responses across repeated runs.

Each model was tested on three prompts and scored manually. Ratings were assigned on a 1–5 scale per criterion and averaged across prompts. Qualitative impressions were recorded, focusing on how well each model translated technical indicators into investment-relevant language.

Responses were analysed for signal grounding, reasoning flow, and verbosity. Models that hallucinated commentary not supported by the input (e.g., "recent news coverage suggests...") were penalised.

### 4.4 Results and Comparison

The results are summarised in the following table:

| Model | Factual Alignment | Interpretability | Stability |
|---|---|---|---|
| mistral-7b | 5/5 | 5/5 | 5/5 |
| phi-7b | 5/5 | 4/5 | 5/5 |
| gemma-7b | 4/5 | 4/5 | 4/5 |
| llama2-7b | 3/5 | 3/5 | 4/5 |
| vicuna-7b | 4/5 | 3/5 | 4/5 |
| codellama-7b | 4/5 | 3/5 | 4/5 |
| neural-chat-7b | 3/5 | 4/5 | 4/5 |
| starling-lm-7b | 4/5 | 4/5 | 3/5 |
| llama2-uncensored-7b | 3/5 | 3/5 | 4/5 |
| llava-7b | 4/5 | 3/5 | 4/5 |
| deepseek-r1-7b | 5/5 | 3/5 | 4/5 |

`mistral-7b` produced consistently high-quality explanations. It maintained clear structure, properly reflected all input values, and varied phrasing across runs. Explanations typically began with a signal summary and concluded with an action justification, closely resembling expert commentary. Its ability to adapt language depending on the action (e.g., cautious phrasing for HOLD or SELL) was particularly effective.

`phi-7b` showed similar strength in factual grounding but was often terse. It used precise language and made no unsupported claims, but the prose was more mechanical. `gemma-7b` was reliable and compact but lacked the fluency of `mistral-7b`.

Other models such as `vicuna-7b` and `starling-lm-7b` produced grammatically correct responses but were more generic. They occasionally repeated template phrasing or omitted one or more key signals (e.g., ignoring RSI or beta). These outputs would require post-processing or filtering before use in production.

### 4.5 Discussion

The comparison highlights how fine-tuned LLM behaviour affects trustworthiness in explainable AI. Models tuned for dialogue or code (`vicuna`, `codellama`) performed adequately but not optimally. Meanwhile, models like `mistral` and `phi`—both tuned for instruction following—demonstrated more appropriate structuring and interpretation.

The scoring rubric also reveals trade-offs. For example, `deepseek` was extremely accurate but slightly verbose and rigid in tone, which might reduce usability for lay audiences. `neural-chat` was fluent but tended toward overconfident or speculative claims.

Stability was assessed via repeated prompt submission. `mistral`, `phi`, and `gemma` retained output meaning while slightly varying sentence structure. This behaviour is desirable: the explanation remains fresh while still grounded.

### 4.6 Conclusion

This evaluation confirms that quantised, open-source 7B LLMs are capable of generating clear and accurate explanations from structured financial inputs. Among the models tested, `mistral-7b` is selected as the default explanation engine. It strikes a compelling balance between fluency, precision, and alignment with non-expert communication needs.

Its performance justifies its use as a core component of the GenAI Advisor system and supports the overall goal of delivering transparent, explainable financial recommendations. The modular design of the prototype allows future integration with upstream pipelines, where generated explanations will be contextualised within retrieved evidence and user-specific preferences.

This work illustrates a practical methodology for evaluating LLMs in structured domains and contributes to the broader discussion on scalable explainability for AI-assisted financial systems.

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